Helpful maths for ML

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Comment 1 – Linear relationships: The equation:

$$y = mx + c,$$

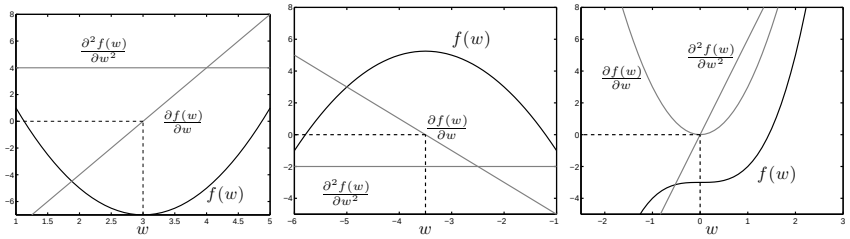
where m and c are constant, defines a linear relationship between x and y. It is called linear because the relationship between x and y could be visualised as a straight line. The following equations are nonlinear due to the more complex forms in which we find the variables x and y:

$$y = mx^2 + c, \quad y = \sin(x), \quad \sqrt{y} = mx + c.$$

The values of m and c to not effect the linearity of the relationship. For example, the following still represent linear relationships between x and y:

$$y = mx + c^2, \quad y = x\sin(m) + c.$$

Comment 2 – Turning points: We can find turning points (that might correspond to minima) of a function, $f(w)$ by searching for points where the gradient of the function, $\frac{\delta f(w)}{\delta w}$ is zero. To determine whether or not a turning point corresponds to a maximum, minimum or saddle point, we can examine the second derivative – $\frac{\delta^2 f(w)}{\delta w^2}$. If, at a turning point $\hat{w}$, the second derivative is positive, we know that this turning point is a minimum. The following 3 plots show three example functions along with their first and second derivatives:



In general, a function may have several turning points. An interesting special case is functions whose second derivative is a positive constant – these correspond to functions that have only one minimum.

Comment 3 – Scalars, vectors and matrices: We will follow the standard convention of representing scalar values by letters (e.g. x), vectors by bold lowercase letters (e.g. $\mathbf{x}$) and matrices by bold uppercase letters (e.g. $\mathbf{X}$). Whilst we shall consistently stick to this notation, different communities have different ways of defining vectors. For example, $\overline{x}$ is common for a vector x.

Comment 4 – Vector transpose: The transpose of a vector $\mathbf{x}$, denoted $\mathbf{x}^\mathsf{T}$, is obtained by rotating the vector such that rather than having one column and several rows, it has one row and several columns. For example:

$$\mathbf{x} = \begin{bmatrix} 4 \\ 7 \\ 11 \\ -2 \end{bmatrix}, \quad \mathbf{x}^\mathsf{T} = [4, 7, 11, -2].$$

Comment 5 – Matrix/vector dimensions and indexing: If we are quoting the size (or the dimension) of a matrix or vector, we give two numbers, starting with the number of rows. For example,

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

has dimension 3×2. A vector is a special case of a matrix where the second dimension is 1. For example,

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$$

could be thought of as a matrix with dimension 4×1.
When indexing elements within a vector, a single number is sufficient (e.g. y_3 for the third element in $\mathbf{y}$ above. When indexing a matrix, we will use two subscripts, starting with the row. For example, a_{21} represents the item in the second row and first column of $\mathbf{A}$ (above). Note that sometimes we will also have a subscript denoting the object index. For example $\mathbf{x}_n$ is the vector holding the nth set of attributes. This index, if present, will always come first. It should be obvious from the context whether or not this index is present.

Comment 6 – Matrix transpose: For a matrix, $\mathbf{X}$, the transpose, $\mathbf{X}^\mathsf{T}$ is formed by turning each row into a column and each column into a row. For example, if $\mathbf{Y} = \mathbf{X}^\mathsf{T}$, then $Y_{ij} = X_{ji}$.

$$\mathbf{X} = \begin{bmatrix} 1 & 4 \\ 3 & 6 \\ -2 & 11 \end{bmatrix}, \quad \mathbf{X}^\mathsf{T} = \begin{bmatrix} 1 & 3 & -2 \\ 4 & 6 & 11 \end{bmatrix}.$$

Comment 7 – Matrix multiplication: To proceed, we must introduce the concept of matrix multiplication. Taking the product, $\mathbf{AB}$ of an $N \times M$ matrix $\mathbf{A}$ and a $P \times Q$ matrix $\mathbf{B}$ is only possible if $M = P$, i.e. the number of columns in $\mathbf{A}$ is equal to the number of rows in $\mathbf{B}$. Assuming that this is the case, the product, $\mathbf{C} = \mathbf{AB}$ is the $N \times Q$ matrix defined such that

$$C_{ij} = \sum_k A_{ik} B_{kj}.$$

It is often helpful to draw the matrices, for example

$$\begin{array}{ccc} & \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix} \\ \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} & \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} & a_{11}b_{13} + a_{12}b_{23} \\ a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} & a_{11}b_{13} + a_{12}b_{23} \end{bmatrix} \end{array}$$

where we can think of computing elements of $\mathbf{C}$ by working simultaneously across the relevant row of $\mathbf{A}$ and column of $\mathbf{B}$.
A special case that we will meet regularly is the *inner product* between two column vectors, defined as $z = \mathbf{x}^\mathsf{T}\mathbf{y}$, the result of which is a scalar. Both vectors must be of the same length and the transpose ensures that the number of columns in $\mathbf{x}$ is the same as the number of rows in $\mathbf{y}$. Applying the same technique as that for matrices, we see that

$$z = \sum_k x_k y_k.$$

Comment 8 – Transpose of a product: The transpose of a matrix product, $(\mathbf{Xw})^\mathsf{T}$ can be expanded by reversing the order of multiplication and transposing the two individual matrices

$$(\mathbf{Xw})^\mathsf{T} = \mathbf{w}^\mathsf{T}\mathbf{X}^\mathsf{T}.$$

To deal with more complex forms, we can apply the same result several times. For example

$$\begin{aligned} (\mathbf{ABCD})^\mathsf{T} &= ((\mathbf{AB})(\mathbf{CD}))^\mathsf{T} \\ &= (\mathbf{CD})^\mathsf{T}(\mathbf{AB})^\mathsf{T} \\ &= \mathbf{D}^\mathsf{T}\mathbf{C}^\mathsf{T}\mathbf{B}^\mathsf{T}\mathbf{A}^\mathsf{T} \end{aligned}$$

Comment 9 – Identity matrix: we will regularly come across the identity matrix $\mathbf{I}_N$. It is the $N \times N$ matrix with ones on the diagonal and zeros elsewhere.

$$\mathbf{I}_1 = 1, \quad \mathbf{I}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{I}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Often, the size of the identity matrix will be obvious from the expression it's found in. In these cases, we will omit the size subscript.
A key property of the identity matrix is that any vector or matrix multiplied by a suitably sized identity matrix is equal to the original matrix or vector. For example, if $\mathbf{y} = [y_1, \ldots, y_D]^\mathsf{T}$ and $\mathbf{I}_D$ is the $D \times D$ identity matrix,

$$\mathbf{y}^\mathsf{T}\mathbf{I}_D = \mathbf{y}, \ \mathbf{I}\mathbf{y} = \mathbf{y}.$$

Similarly, for an $N \times M$ matrix,

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1M} \\ a_{21} & a_{22} & \ldots & a_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ a_{N1} & a_{N2} & \ldots & a_{NM} \end{bmatrix}$$

$$\mathbf{A}\mathbf{I}_M = \mathbf{A}, \ \mathbf{I}_N\mathbf{A} = \mathbf{A}.$$

Multiplying a scalar by an identity results in a matrix with the scalar value on each diagonal element. An example that crops up a lot is:

$$\sigma^2\mathbf{I}_M = \begin{bmatrix} \sigma^2 & 0 & \ldots & 0 \\ 0 & \sigma^2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \sigma_2 \end{bmatrix}$$

Comment 10 – Matrix inverse: The inverse of a matrix $\mathbf{A}$ is defined as the matrix, $\mathbf{A}^{-1}$ that satisfies $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$. We don't provide the general form for inverting a matrix here, but from school mathematics, a 2×2 matrix can be inverted with the following formula:

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \mathbf{A}^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

A special case that we will come across regularly is the inverse of a matrix that only has values on the diagonal (i.e. all off-diagonal elements are zero). The inverse of such a matrix is another diagonal matrix where each diagonal element is simply the inverse of the corresponding element in the original. For example,

$$\mathbf{A} = \begin{bmatrix} a_{11} & 0 & \ldots & 0 \\ 0 & a_{22} & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & a_{DD} \end{bmatrix}, \ \mathbf{A}^{-1} = \begin{bmatrix} a_{11}^{-1} & 0 & \ldots & 0 \\ 0 & a_{22}^{-1} & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & a_{DD}^{-1} \end{bmatrix}$$

It is worth noting that this definition implies that the inverse of an identity matrix is simply another identity matrix:

$$\mathbf{I}^{-1} = \mathbf{I}.$$

Comment 11 – Conditional probabilities and marginalisation – an example: Lets assume that we have a fair coin and two dice (one of which is a little unusual). We will generate a coin toss (X) and a dice roll (Y) using the following procedure. Firstly, toss the coin. If it gives heads, roll die 1. If it gives tails, roll die 2. Die 1 and die 2 are different, with probabilities defined in the following table

	1	2	3	4	5	6	
Die 1	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$= P(y\|X = H)$
Die 2	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{6}$	$\frac{1}{12}$	$= P(y\|X = T)$

So, the probability of rolling say a 3 is 1/6 with die 1 and 1/4 with die 2. As we roll die 1 if our coin showed heads and dice 2 if tails, we have the following conditional distributions

$$P(y|X = H),\ P(y|X = T),$$

i.e. the distribution over Y depends on the outcome of X. The joint distribution is given as (Equation **??**):

$$p(y, x) = p(y|x)p(x).$$

We can use this to compute the probability of rolling a 3 *and* a head:

$$P(Y = 3, x = H) = P(Y = 3|X = H)P(X = H) = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12}.$$

Alternatively, a 3 *and* a tail

$$P(Y = 3, X = T) = P(Y = 3|X = H)P(X = T) = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}.$$

Perhaps more interestingly, we can compute the marginal distribution for Y. From our definition (Equation **??**):

$$P(y) = \sum_x P(y, x) = \sum_x P(y|x)P(x).$$

Therefore, the probability of rolling a 3 is:

$$\begin{aligned} P(Y = 3) &= \sum_x P(Y = 3|x)P(x) \\ &= P(Y = 3|X = H)P(X = H) + P(Y = 3|X = T)P(X = T) \\ &= \frac{1}{6} \times \frac{1}{2} + \frac{1}{4} \times \frac{1}{2} = \frac{5}{24}. \end{aligned}$$

Comment 12 – Vector random variables: It will often be necessary to define probability distributions over vectors. This is nothing more than a shorthand way of defining large joint distributions. For example, the values that could be taken on by random variables $X_1, X_2, \ldots, X_N$ can be expressed as the vector $\mathbf{x} = [x_1, x_2, \ldots, x_N]^\mathsf{T}$. Using this shorthand:

$$p(\mathbf{x}) = p(x_1, x_2, \ldots, x_N) = P(X_1 = x_1, X_2 = x_2, \ldots, X_N = x_N).$$

Even though $\mathbf{x}$ is a vector, $p(\mathbf{x})$ is a scalar quantity, just as $P(X_1 = x_1, X_2 = x_2, \ldots, X_N = x_N)$ is.

Comment 13 – Combinations: N choose y, written as

$$\left(\begin{array}{c} N \\ y \end{array} \right)$$

is mathematical shorthand for the number of ways in which y distinct objects can be chosen from a set of N objects. For example, $\left(\begin{array}{c} 4 \\ 1 \end{array} \right)$ would be 4 – there are 4 ways I can choose one object from four objects – object 1 on its own, object 2 on its own, object 3 on its own or object 4 on its own. $\left(\begin{array}{c} 4 \\ 2 \end{array} \right)$ is 6 – the possible choices are 1 and 2, 1 and 3, 1 and 4, 2 and 3, 2 and 4 or 3 and 4. In general,

$$\left(\begin{array}{c} N \\ y \end{array} \right) = \frac{N!}{y!(N-y)!}$$

where $N!$ (read N factorial) is

$$\prod_{i=1}^{N} i = N \times (N-1) \times (N-2) \times \ldots \times 1.$$

Comment 14 – Definite Integrals: When differentiating a function including a constant term, the term disappears, e.g.

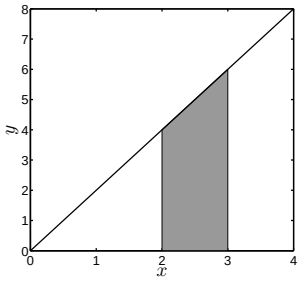
$$\frac{d}{dx}(x^2 + 3) = 2x.$$

Hence, when we are integrating a function, we have to admit the possibility that there might be a constant term

$$\int 2x \; dx = x^2 + C.$$

This is called an indefinite integral as we don't know the value of C.

Often we will be interested in using integration to compute the area under a curve. For example, here we are interested in computing the area under the curve $y = 2x$ between $x = 2$ and $x = 3$ as shown int the plot on the right. This is calculated as



$$\int_2^3 2x \; dx = [x^2 + C]_2^3$$

where the $[\cdot]_a^b$ means take the value of the object inside the brackets when $x = a$ away from the value when $x = b$. In this case, this suggests

$$(3^2 + C) - (2^2 + C) = 9 - 4 + C - C = 5.$$

This is a definite integral – the constants cancel out and the answer is exact.

Comment 15 – Matrix determinant: The determinant of a square matrix, denoted $|\mathbf{A}|$ for matrix $\mathbf{A}$ is a useful quantity, especially when dealing with multivariate Gaussians. For large matrices, it is too cumbersome to calculate by hand but it can be done for small matrices. For example, for a 2×2 matrix

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \ |\mathbf{A}| = ad - bc,$$

but for anything bigger than this it is safest to resort to a computer unless the matrix has special structure. One special matrix that we will see a lot of is a square matrix that only has diagonal elements (all off-diagonal elements are zero). In this case, the determinant is simply the product of these elements. For example,

$$\mathbf{A} = \begin{bmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{DD} \end{bmatrix}, \ |\mathbf{A}| = \prod_{d=1}^{D} a_{dd}.$$

It is not easy to gain an intuition into what the determinant represents. Its role in the normalisation constant of the multivariate Gaussian leads us to think of it as related to the volume of the Gaussian unnormalised Gaussian (remember that the normalised volume must be equal to 1) and it may be useful to think of it in this way.

Comment 16 – Hessian matrix: A Hessian matrix is square matrix containing all of the second-order partial derivatives of a function. For example, the Hessian matrix for a function $f(\mathbf{x}; \mathbf{w})$ with parameters $\mathbf{w} = [w_1, \dots, w_K]^T$ would be

$$\mathbf{H} = \begin{bmatrix} \frac{\partial^2 f}{\partial w_1^2} & \frac{\partial^2 f}{\partial w_1 \partial w_2} & \dots & \frac{\partial^2 f}{\partial w_1 \partial w_K} \\ \frac{\partial^2 f}{\partial w_2 \partial w_1} & \frac{\partial^2 f}{\partial w_2^2} & \dots & \frac{\partial^2 f}{\partial w_2 \partial w_K} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial w_K \partial w_1} & \frac{\partial^2 f}{\partial w_K \partial w_2} & \dots & \frac{\partial^2 f}{\partial w_K^2} \end{bmatrix}.$$

We can use the Hessian to tell us something about turning points in $f(\mathbf{x}; \mathbf{w})$. For example, if the Hessian is *negative definite* at some turning point $\mathbf{w}\mathbf{w}^\mathsf{T}$ then we know that that turning point corresponds to a maximum.

Comment 17 – Negative definite matrices: A real-valued matrix $\mathbf{H}$ is negative definite if

$$\mathbf{x}^\mathsf{T}\mathbf{H}\mathbf{x} < 0$$

for all vectors of real values $\mathbf{x}$.

Comment 18 – Matrix trace: The trace of a square matrix $\mathbf{A}$, denoted $\mathsf{Tr}(\mathbf{A})$ is the sum of the diagonal elements of $\mathbf{A}$. For example if

$$\mathbf{A} = \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1D} \\ A_{21} & A_{22} & \cdots & A_{2D} \\ \vdots & \vdots & \ddots & \vdots \\ A_{D1} & A_{D2} & \cdots & A_{DD} \end{bmatrix}$$

Then

$$\mathsf{Tr}(\mathbf{A}) = \sum_{d=1}^{D} A_{dd}.$$

It follows that if $\mathbf{A} = \mathbf{I}_D$, i.e. the $D \times D$ identity matrix,

$$\mathsf{Tr}(\mathbf{I}_D) = \sum_{d=1}^{D} 1 = D.$$

A useful identity that we will often use is that

$$\mathsf{Tr}(\mathbf{AB}) = \mathsf{Tr}(\mathbf{BA}).$$

Also, the trace of a scalar is just equal to the scalar value (a scalar could be thought of as a 1×1 matrix), i.e.

$$\mathsf{Tr}(a) = a.$$

or, if $\mathbf{w} = [w_1, \ldots, w_D]^\mathsf{T}$,

$$\mathsf{Tr}(\mathbf{w}^\mathsf{T}\mathbf{w}) = \mathbf{w}^\mathsf{T}\mathbf{w}$$

because the result of $\mathbf{w}^\mathsf{T}\mathbf{w}$ is a scalar.

Comment 19 – Conjugate priors: A likelihood-prior pair is said to be conjugate if they result in a posterior which is of the same form as the prior. This enables us to compute the posterior density analytically without having to worry about computing the denominator in Bayes' rule, the marginal likelihood. Some common conjugate pairs are listed in the table to the right.

Prior	Likelihood
Gaussian	Gaussian
Beta	Binomial
Gamma	Gaussian
Dirichlet	Multinomial

Comment 20 – The Newton-Raphson method: The Newton-Raphson method (also known as the Newton method) is a general method for finding points where functions are equal to zero; i.e., finding points where the function $f(x) = 0$. Given a current estimate of the zero point: x_n, we update it by moving to the point where the tangent to the function at x_n passes through the x-axis. This point can be computed by approximating the gradient as a change in $f(x)$ divided by a change in x. Defining $\partial f(x)/\partial x$ as $f'(x)$:

$$\begin{aligned} f'(x_n) &= \frac{f(x_n) - 0}{x_n - x_{n+1}} \\ (x_n - x_{n+1}) f'(x_n) &= f(x_n) \\ x_{n+1} &= x_n - \frac{f(x_n)}{f'(x_n)}. \end{aligned}$$

The method can also be used to find minima and maxima as these are simply points where the gradient passes through zero. Therefore, we simply replace $f(x)$ with its derivative $f'(x)$ and $f'(x)$ with its derivative $f''(x)$:

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}.$$

This is readily extendable to functions of a vector – say $\mathbf{x}$. In this instance, $f'(x_n)$ is replaced by the vector of partial derivatives evaluated at $\mathbf{x}_n$ and $1/f''(x_n)$ is replaced by the inverse of the Hessian matrix – $\partial^2 f(\mathbf{x})/\partial \mathbf{x} \partial \mathbf{x}^\mathsf{T}$ – evaluated at $\mathbf{x} = \mathbf{x}_n$.

Comment 21 – The chain rule : When taking partial derivatives, it is often convenient to use the chain rule. The chain rule states that:

$$\frac{\partial f(g(\mathbf{w}))}{\partial \mathbf{w}} = \frac{\partial f(g(\mathbf{w}))}{\partial g(\mathbf{w})} \frac{\partial g(\mathbf{w})}{\partial \mathbf{w}}.$$

As an example, let:

$$f(\mathbf{w}) = t_n \log P_n$$

where

$$P_n = \frac{1}{1 + \exp(-\mathbf{w}^\mathsf{T} \mathbf{x}_n)}.$$

To compute $\frac{\partial f(\mathbf{w})}{\partial \mathbf{w}}$, we can use the chain rule as follows:

$$\frac{\partial f(\mathbf{w})}{\partial \mathbf{w}} = \frac{\partial f(\mathbf{w})}{\partial P_n} \frac{\partial P_n}{\partial \mathbf{w}} = \frac{t_n}{P_n} \frac{\partial P_n}{\partial \mathbf{w}}.$$

Comment 22 – Taylor expansions: The Taylor expansion is a way of approximating a function. The approximation is always made 'about' some value – the approximation will tend to diverge from the true function as we move away from that value. The definition of the Taylor series of $f(w)$ about $\widehat{w}$ is:

$$f(w) = \sum_{n=0}^{\infty} \frac{(w-\widehat{w})^n}{n!} \left.\frac{\partial^n f(w)}{\partial w^n}\right|_{\widehat{w}}$$

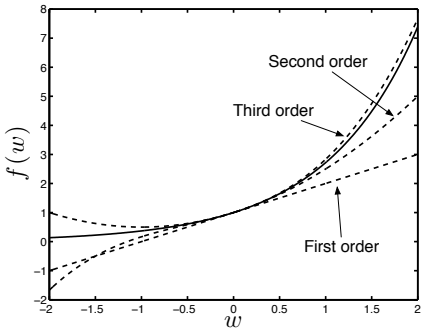
where $\left.\frac{\partial^n f(w)}{\partial w^n}\right|_{\widehat{w}}$ is the nth derivative of $f(w)$ with respect to w, evaluated at $\widehat{w}$. When $n = 0$, this derivative is simply the function $f(w)$. If we only compute a finite number of terms, we will have an approximation to the function. A first order approximation would just include terms $n = 0$ and $n = 1$ – an nth order approximation includes all terms up to and including term n. For example, we can approximate $f(w) = \exp(w)$ at $\widehat{w} = 0$:

$$\exp(w) = \exp(\widehat{w}) + \frac{w}{1!}\exp(\widehat{w}) + \frac{w^2}{2!}\exp(\widehat{w}) + \ldots$$

Now, $\exp(\widehat{w}) = 1$, so:

$$\exp(w) = 1 + \frac{w}{1!} + \frac{w^2}{2!} + \frac{w^3}{3!} + \ldots$$

The approximation will get better and better as we add more and more terms. This can be seen in the figure on the right.



Comment 23 – Constrained optimisation with Lagrange multipliers: At various points in this book we will need to perform constrained optimisations – finding the values of a set of parameters that maximises (or minimises) an objective function but that also satisfy some constraints. This can be done using *Lagrange* multipliers. In particular, we make a new objective function which includes the original plus an additional term for each constraint. The form of these terms is chosen such that the optimum of the new function is equal to the optimum of the constrained problem.

For example, suppose we wish to minimise $f(w)$ subject to the constraint $g(w) \leq a$:

$$\begin{aligned} \underset{w}{\text{argmin}} \quad & f(w) \\ \text{subject to} \quad & g(w) \leq a \end{aligned}$$

The new objective function is produced by adding a Lagrangian term of the form $\lambda(a - g(w))$ and optimised over both w *and* the Lagrange multiplier λ:

$$\begin{aligned} \underset{w,\lambda}{\text{argmin}} \quad & f(w) - \lambda(g(w) - b) \\ \text{subject to} \quad & \lambda > 0. \end{aligned}$$

We are not going to go into the details of how this works here. Whenever we perform constrained optimisation, we will state the necessary Lagrangian terms without any further details. For more details, see the suggested reading at the end of this chapter.

Comment 24 – The product rule of differentiation: The product rule is used when we need to differentiate a product of two functions of the same variable with respect to the variable. For example, if:

$$f(a) = g(a)h(a),$$

the product rule states that:

$$\frac{\partial f(a)}{\partial a} = g(a)\frac{\partial h(a)}{\partial a} + \frac{\partial g(a)}{\partial a}h(a).$$

For example, to differentiate $a \log a$ with respect to a, this gives:

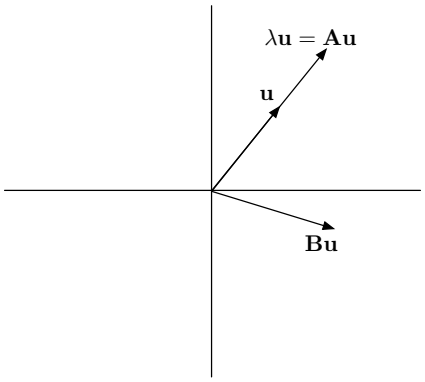
$$a \times \frac{1}{a} + 1 \times \log(a) = 1 + \log(a).$$

Comment 25 – Eigenvectors and eigen values: The eigenvector/eigenvalue equation for some square matrix $\mathbf{A}$ is given as:

$$\lambda_i \mathbf{u}_i = \mathbf{A}\mathbf{u}_i. \tag{1}$$

The solutions to this equation are pairs of eigenvalues (λ_i) and eigenvectors ($\mathbf{u}_i$).

The figure on the right provides some intuition for this equation. Multiplying an M-dimensional vector $\mathbf{u}$ by an $M \times M$ matrix $\mathbf{B}$ results in another M-dimensional vector. Therefore, we can consider the matrix $\mathbf{B}$ as defining a rotation of the vector $\mathbf{u}$. Different $\mathbf{B}$ matrices will produce different rotations. The solutions to Equation **??**, for a particular matrix $\mathbf{A}$ are the vectors $\mathbf{u}$ for which applying the rotation $\mathbf{A}$ only results in a change in the length of $\mathbf{u}$. The magnitude of this change is given by the scalar λ.



In general, if the matrix $\mathbf{A}$ has M rows and M columns, there are M eigenvector/eigenvalue pairs that solve Equation 1. The M eigenvectors will be orthogonal. We are not going to go into the detail of how to solve the eigenvalue/eigenvector equation. Routines for doing it are common, for example, the `eigs` function in Matlab.

Comment 26 – Kullback-Leibler divergence: It is often important to be able to quantify the difference between two probability distributions. For example, if we are trying to find an approximate posterior that is similar to the true posterior, we need to define what we mean by similar! The Kullback-Leibler divergence is one such quantity that appears in the derivation of Variational Bayesian techniques. It is defined for discrete and continuous distributions as:

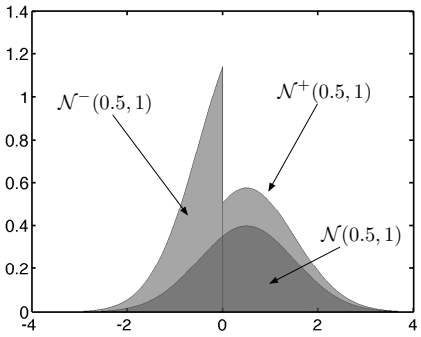
$$\begin{aligned} \mathrm{KL}[q(x)||p(x)] &= \int q(x) \log \frac{p(x)}{q(x)}\, dx \quad \text{(continuous)} \\ \mathrm{KL}[q(x)||p(x)] &= \sum_x q(x) \log \frac{p(x)}{q(x)} \quad \text{(discrete)} \end{aligned}$$

For continuous distributions it is almost always intractable to compute due to the integral over a potentially high-dimensional space.

An important property of the KL-divergence is its asymmetry – $\mathrm{KL}[q(x)||p(x)] \neq \mathrm{KL}[p(x)||q(x)]$. KL divergence is always less than or equal to zero with the maximum value of zero being reached when $p(x) = q(x)$.

Comment 27 – Truncated Gaussian densities: A truncated Gaussian density is a Gaussian density with an additional restriction placed on the random variable. We will only be interested in Gaussian densities truncated above or below the origin.

The figure on the right shows a standard Gaussian density (with mean 0.5 and variance 1) as well as the positively and negatively truncated densities. The truncated densities have the same shape as the standard density but are both higher. This is because they must still integrate to 1 over their reduced range.



Sampling from a truncated Gaussian is fairly straightforward – one can just sample from the un-truncated density and throw away samples that do not fulfil the necessary constraint. The expected value of positively and negatively truncated Gaussians are given by:

$$p(x) = \mathcal{N}^+(\mu, \sigma^2), \qquad \langle x \rangle = \mu + \frac{\mathcal{N}_{\mu/\sigma}(0,1)}{1 - \phi(-\mu/\sigma)}$$

$$p(x) = \mathcal{N}^-(\mu, \sigma^2), \qquad \langle x \rangle = \mu - \frac{\mathcal{N}_{\mu/\sigma}(0,1)}{\phi(-\mu/\sigma)}$$

where $\mathcal{N}_a(0,1)$ is the standard Gaussian pdf evaluated at a and $\phi(a)$ is the standard normal cdf function evaluated at a.